Achouri Youcef Islem

Software Engineering and Spatial Computing Internship

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Final-year engineering student building interactive 3D and AI applications. Available January 2027.

EDUCATION

ESI-SBA 2021 - Present

École Nationale Supérieure d'Informatique de Sidi Bel Abbès
State Engineering Degree in Computer Science - fifth year, degree in progress.
Specialty: Artificial Intelligence and Data Structures (AIDS).

SELECTED PROJECTS

Spatial Computer Vision | Python, PyTorch, Babylon.js

- Built a pipeline that segments windows and doors in panoramic imagery using Mask2Former and SAM, then projects those masks onto 3D wall geometry.
- Implemented ray-plane lifting and depth checks for occlusion, with a WebGPU viewer for inspecting reconstructed scenes and lighting.

Conversational Learning Platform | React, Vapi, Supabase 2025

- Built a voice tutor interface connecting WebRTC audio with Deepgram speech recognition, GPT-4 dialogue and ElevenLabs speech synthesis.
- Implemented tutor configuration, live transcript events and speech-state feedback, with Clerk authentication and Supabase session metadata.

Arabic Structured Information Extraction | Python, Qwen 2.5, Ollama 2025

- Authored the NLP module for the team's PlanMilt DZ academic GIS project, converting Arabic descriptions into structured scene data.
- Defined the extraction schema and prompts, with JSON parsing, validation and synthetic Arabic examples for local evaluation.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, C++.
Application development: React, FastAPI, NestJS, Django, PostgreSQL, MinIO.
Spatial computing: Babylon.js, Three.js, WebGPU, LiDAR point clouds.
Delivery and testing: Docker, Docker Compose, Linux, Git, pytest.

LEADERSHIP AND COMMUNICATION

Vice President | INGENIUMS Club, ESI-SBA 2023 - 2024

- Co-organized student technology events and programming contests, including Ingeneer, Ingehack and Ingecode.
- Spoke at ESI-SBA Welcome Day and demonstrated technical projects at campus exhibitions.

AWARDS

2nd place, DevFest Hackathon 2023 - GDG Sidi Bel Abbès, Team Infinity.
3rd place, NCS Hackathon 2024 - Numidia Computer Society / NiT.

Languages: English and French (strong proficiency).